BACKPLANE

Backplane

File: backplane/backplane.kicad_sch

FLUXO DE DADOS

PRANCHA I: PLACA DOS REGISTRADORES

OK

Placas de registradores:
* FR1-1 (bits pares)
* FR2-2 (bits ímpares)

File: FR1-1/FR1-1.kicad_sch

PRANCHA II: CIRCUITO - UNIDADE ARITMÉTICA

OK

Placas de somador:
* FS1-3 (bits menos significativos)
* FS2-4 (bits mais significativos)

File: FS1-3/FS1-3.kicad_sch

PRANCHA III: ACUMULADOR

WIP

Placa de acumulador:
* FAC-5

File: FAC-5/FAC-5.kicad_sch

PRANCHA IV: FLUXO DE DADOS

OK

Placa:
* FM1-6 (mais-um e porta de seleção 6)

File: FM1-6/FM1-6.kicad_sch

PRANCHA V: PLACA CONTROLE DA MEMÓRIA

OK

Placa:
* FCM-7 (interface com memória)

File: FCM-7/FCM-7.kicad_sch

PRANCHA VI: DADOS PARA CARTÕES INTERFACE

OK

Placa:
* FIN-8 (sinais para cartões de interface)

File: FIN-8/FIN-8.kicad_sch

PAINEL

PRANCHA VII: PLACA MOSTRA REGISTRADORES

OK

Placa:
* PMR-1 (drivers para os LEDs)

File: PMR-1/PMR-1.kicad_sch

PRANCHA VIII: PLACA CHAVES E PILOTO

OK

Placa:
* PCP-2 ("debounce" para chaves
e drivers para lâmpadas piloto)

File: PCP-2/PCP-2.kicad_sch

UNIDADE DE CONTROLE

PRANCHA IX: RELÓGIO CENTRAL

OK

Placa:
* CRC-1

File: CRC-1/CRC-1.kicad_sch

PRANCHA X: DECODIFICADOR

OK

Placa:
* CDE-3

File: CDE-3/CDE-3.kicad_sch

PRANCHA XI: CONTROLE DE ESTADO

W.I.P.

Placa:
* CCE-2

File: CCE-2/CCE-2.kicad_sch

PRANCHA XII: PLACA CONTROLE 1

OK

Placa:
* CT1-4

File: CT1-4/CT1-4.kicad_sch

PRANCHA XIII: PLACA SINAIS CONTROLE

OK

Placa:
* CT2-5

File: CT2-5/CT2-5.kicad_sch

PRANCHA XIV: PLACA DE CONTROLE

OK

Placa:
* CT3-6

File: CT3-6/CT3-6.kicad_sch

PRANCHA XV: SINAIS DE CONTROLE

OK

Placa:
* CT4-7

File: CT4-7/CT4-7.kicad_sch

PRANCHA XVI: SINAIS DE CONTROLE

OK

Placa:
* CT5-8

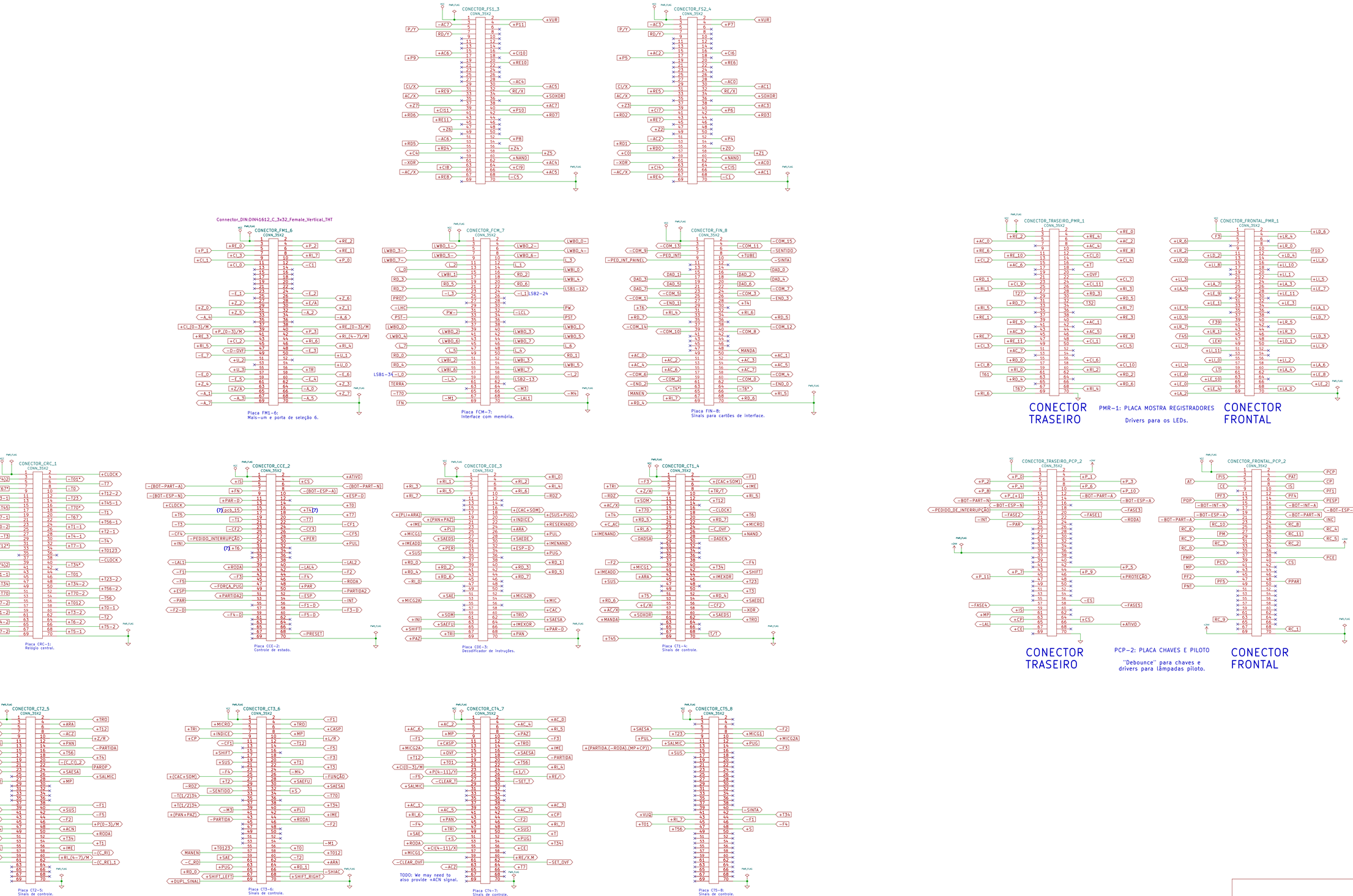
File: CT5-8/CT5-8.kicad_sch

Re-desenho: Felipe Sanches
Projeto: Edson Fregni
Sinais de Controle
Computador de Controle - Patinho Feio
EPUSP LSD

Sheet: /
File: patinhofeio.kicad_sch

Title: PRANCHA XVI

Size: A4	Date: 2 aug 2016	**Rev:**
KiCad E.D.A. 10.0.3		Id: 1/18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CONN_35X2
CONECTOR_FS2_4
CONN_35X2
Connector_DIN:DIN41612_C_3x32_Female_Vertical_THT
CONECTOR_FM1_6
CONN_35X2
Placa FM1-6:
Mais-um e porta de seleção 6.
CONECTOR_FCM_7
CONN_35X2
Placa FCM-7:
Interface com memória.
CONECTOR_FIN_8
CONN_35X2
Placa FIN-8:
Sinais para cartões de interface.
CONECTOR_TRASEIRO_PMR_1
CONN_35X2
CONECTOR TRASEIRO
PMR-1: PLACA MOSTRA REGISTRADORES
Drivers para os LEDs.
CONECTOR_FRONTAL_PMR_1
CONN_35X2
CONECTOR FRONTAL
CONECTOR_CRC_1
CONN_35X2
Placa CRC-1:
Relógio central.
CONECTOR_CCE_2
CONN_35X2
Placa CCE-2:
Controle de estado.
CONECTOR_CDE_3
CONN_35X2
Placa CDE-3:
Decodificador de instruções.
CONECTOR_CT1_4
CONN_35X2
Placa CT1-4:
Sinais de controle.
CONECTOR_TRASEIRO_PCP_2
CONN_35X2
CONECTOR TRASEIRO
PCP-2: PLACA CHAVES E PILOTO
"Debounce" para chaves e drivers para lâmpadas piloto.
CONECTOR_FRONTAL_PCP_2
CONN_35X2
CONECTOR FRONTAL
CONECTOR_CT2_5
CONN_35X2
Placa CT2-5:
Sinais de controle.
CONECTOR_CT3_6
CONN_35X2
Placa CT3-6:
Sinais de controle.
CONECTOR_CT4_7
CONN_35X2
TODO: We may need to
also provide +ACN signal.
Placa CT4-7:
Sinais de controle.
CONECTOR_CT5_8
CONN_35X2
Placa CT5-8:
Sinais de controle.
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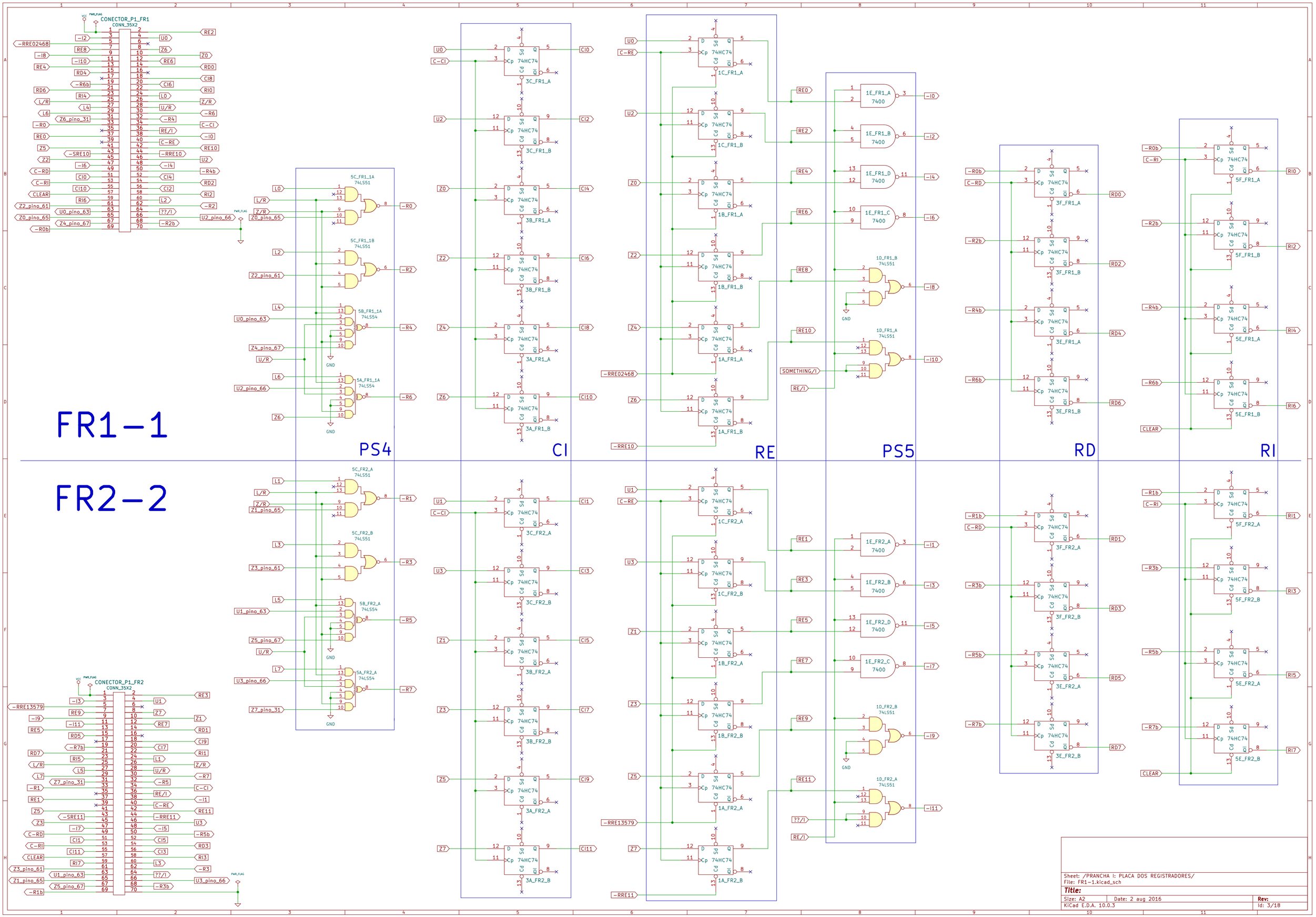
Size: A1
Date:
Rev:
KiCad E.D.A. 10.0.3
Id: 2/18


FR1-1
FR2-2
CONECTOR_P1_FR1
CONN_35X2
CONECTOR_P1_FR2
CONN_35X2
PS4
CI
RE
PS5
RD
RI
5C_FR1_1A 74LS51
5C_FR1_1B 74LS51
5B_FR1_1A 74LS54
5A_FR1_1A 74LS54
5C_FR2_A 74LS51
5C_FR2_B 74LS51
5B_FR2_A 74LS54
5A_FR2_A 74LS54
74HC74
7400
1D_FR1_B 74LS51
1D_FR1_A 74LS51
1D_FR2_B 74LS51
1D_FR2_A 74LS51
GND
PWR_FLAG
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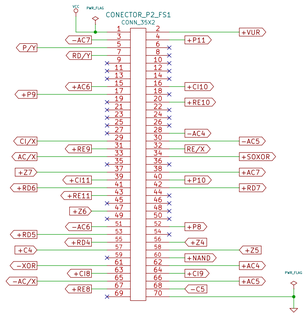
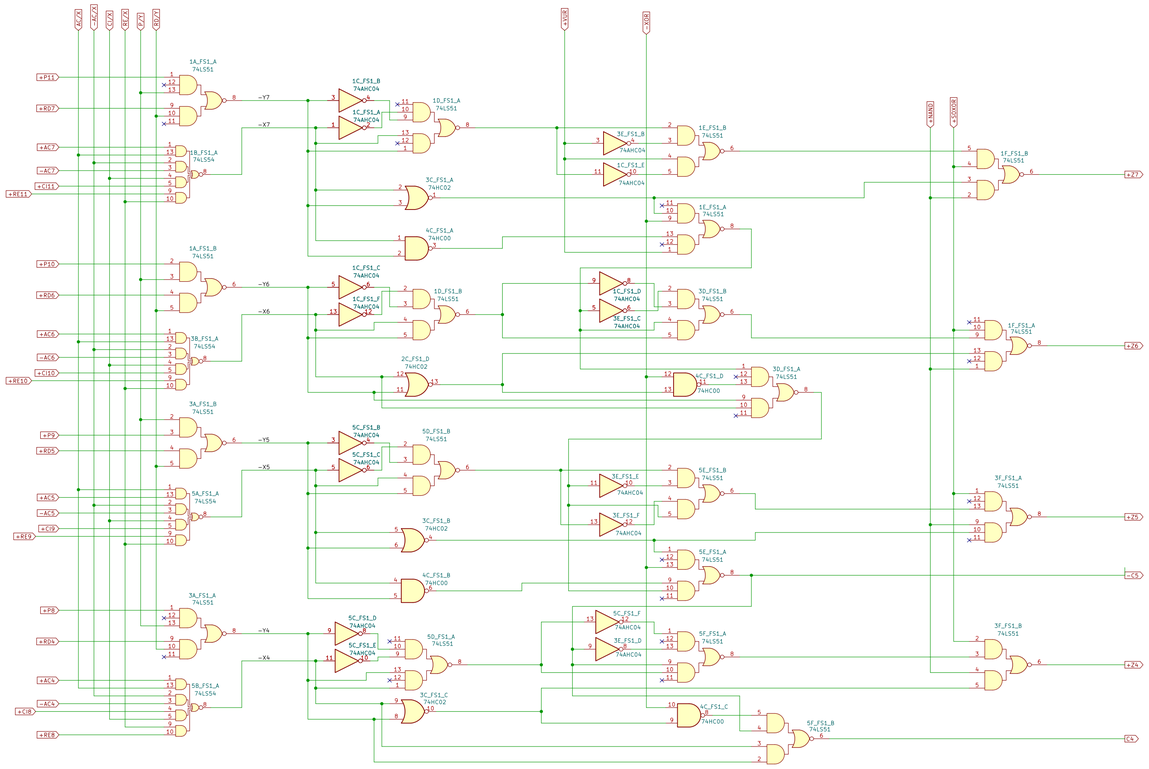
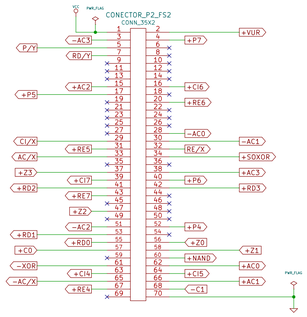
Size: A2
Date: 2 aug 2016
KiCad E.D.A. 10.0.3
Rev:
Id: 3/18







Sheet: /PRANCHA II: CIRCUITO – UNIDADE ARITMÉTICA/
File: FS1-3.kicad_sch

Title:

Size: A1	Date: 2026-06-29	Rev:
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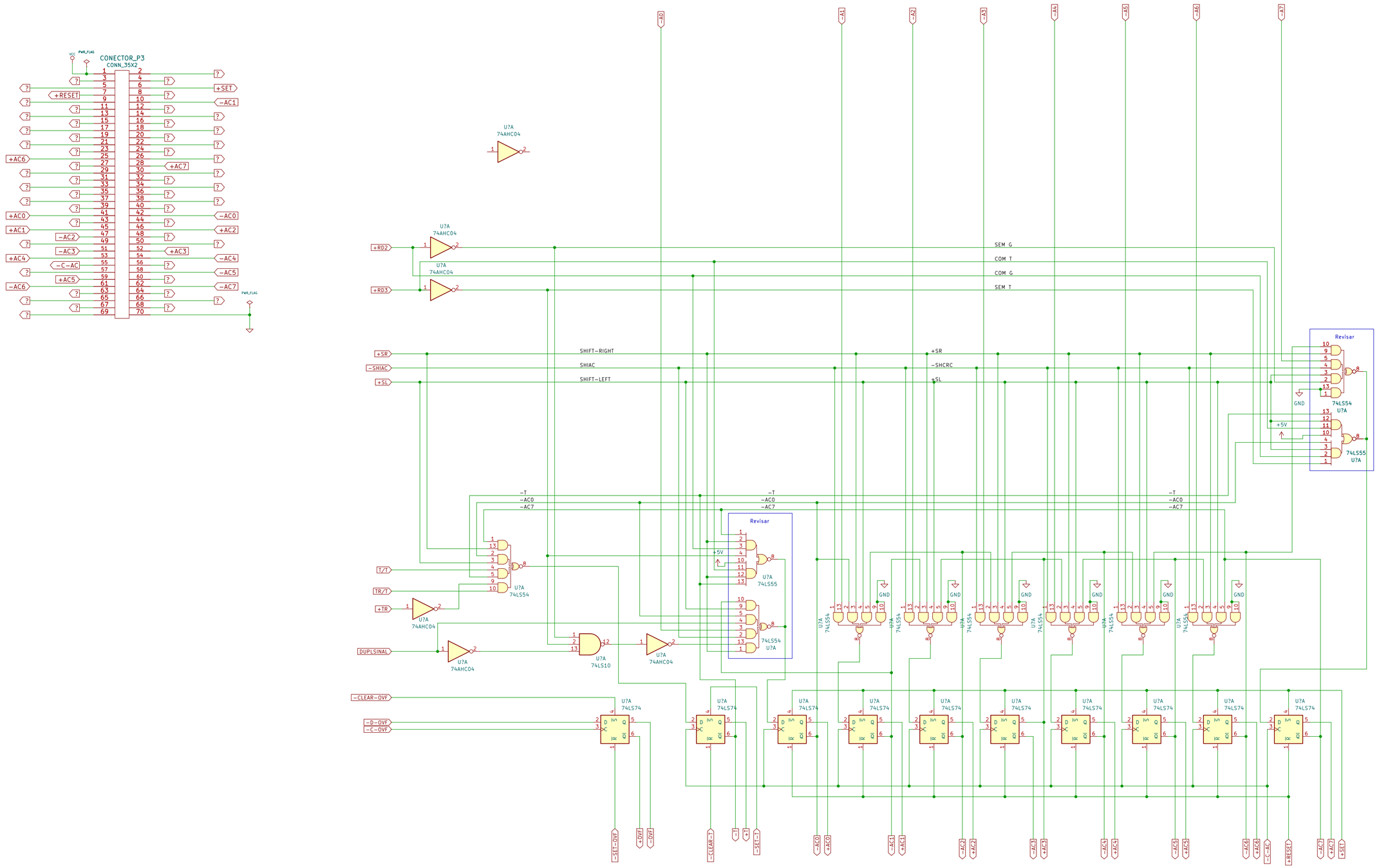
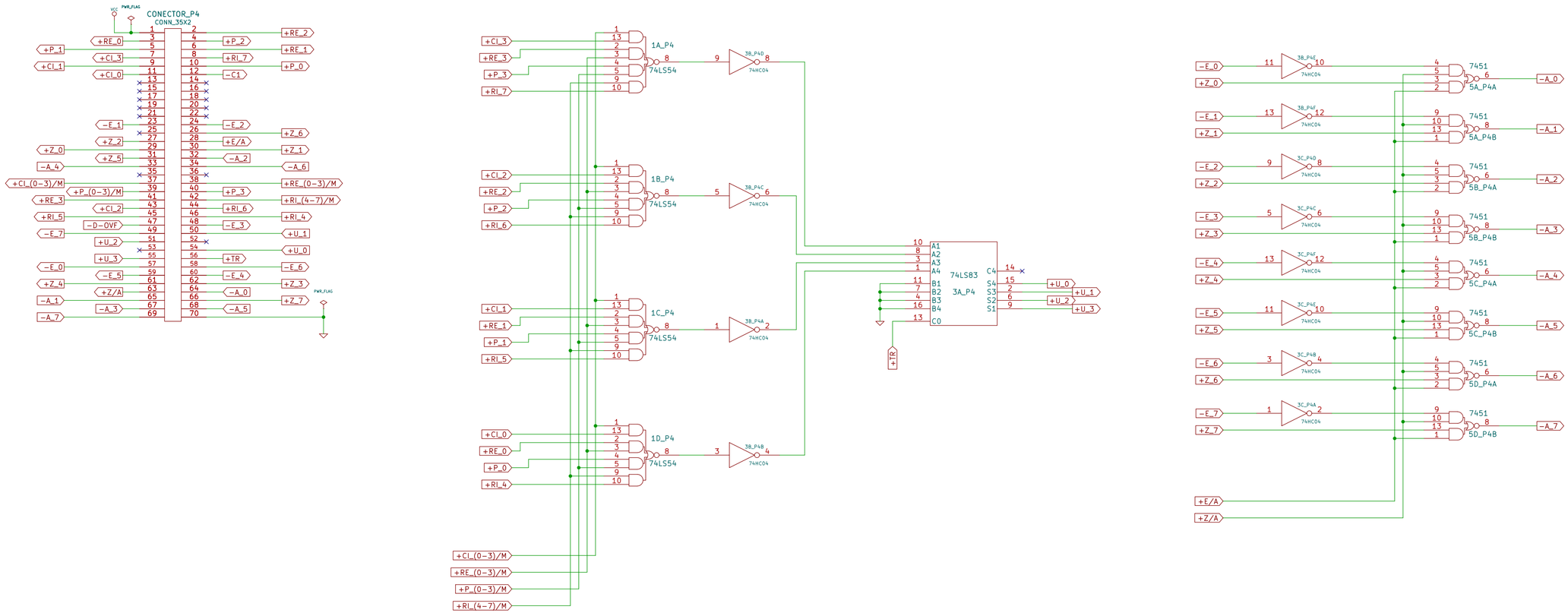


Sheet: /PRANCHA III: ACUMULADOR/
File: FAC-5.kicad_sch

Title:

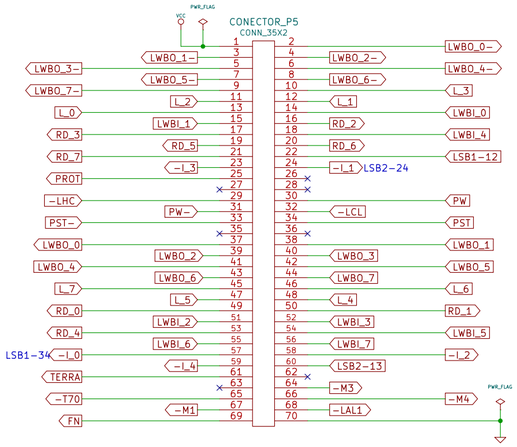
Size: A2	Date: 2 aug 2016	Rev:
KiCad E.D.A. 10.0.3		Id: 5/18

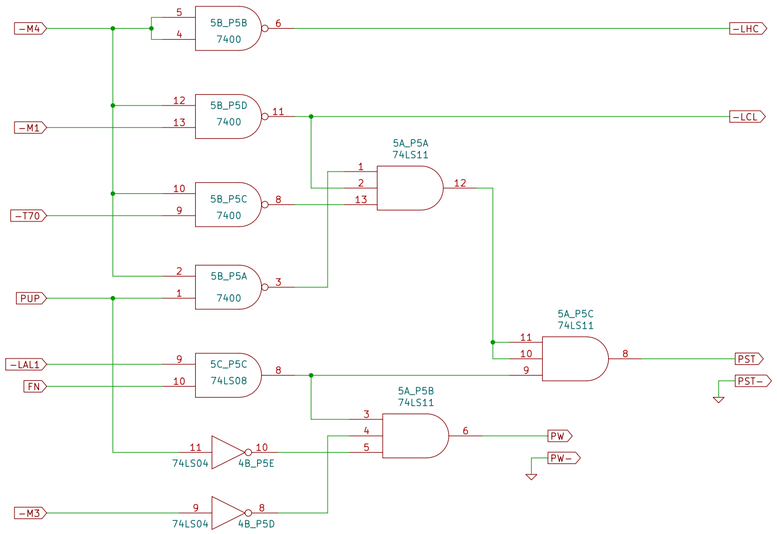


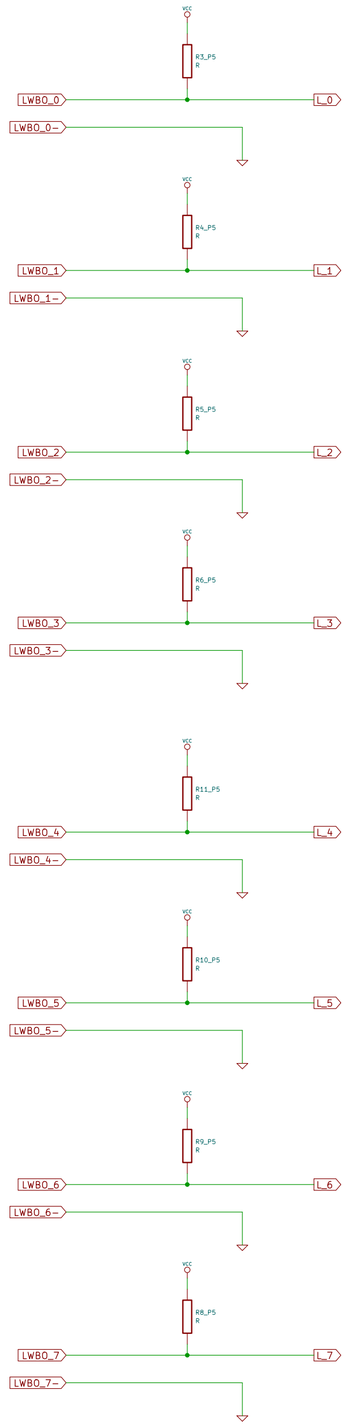
Sheet: /PRANCHA IV: FLUXO DE DADOS/
File: FM1-6.kicad_sch

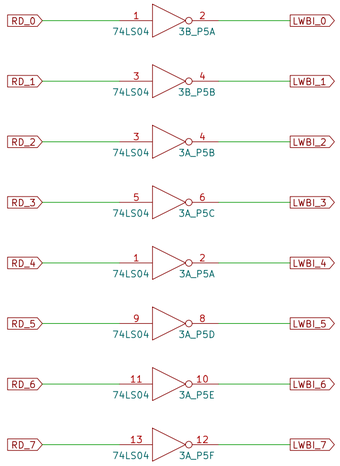
Title: MAIS UM E PORTA DE SELEÇÃO

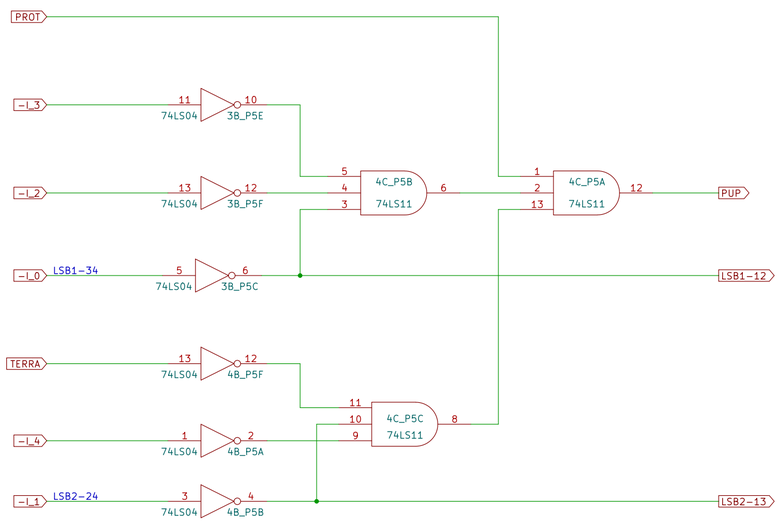
Size: A2	Date: 2 aug 2016	Rev:
KiCad E.D.A. 10.0.3		Id: 6/18







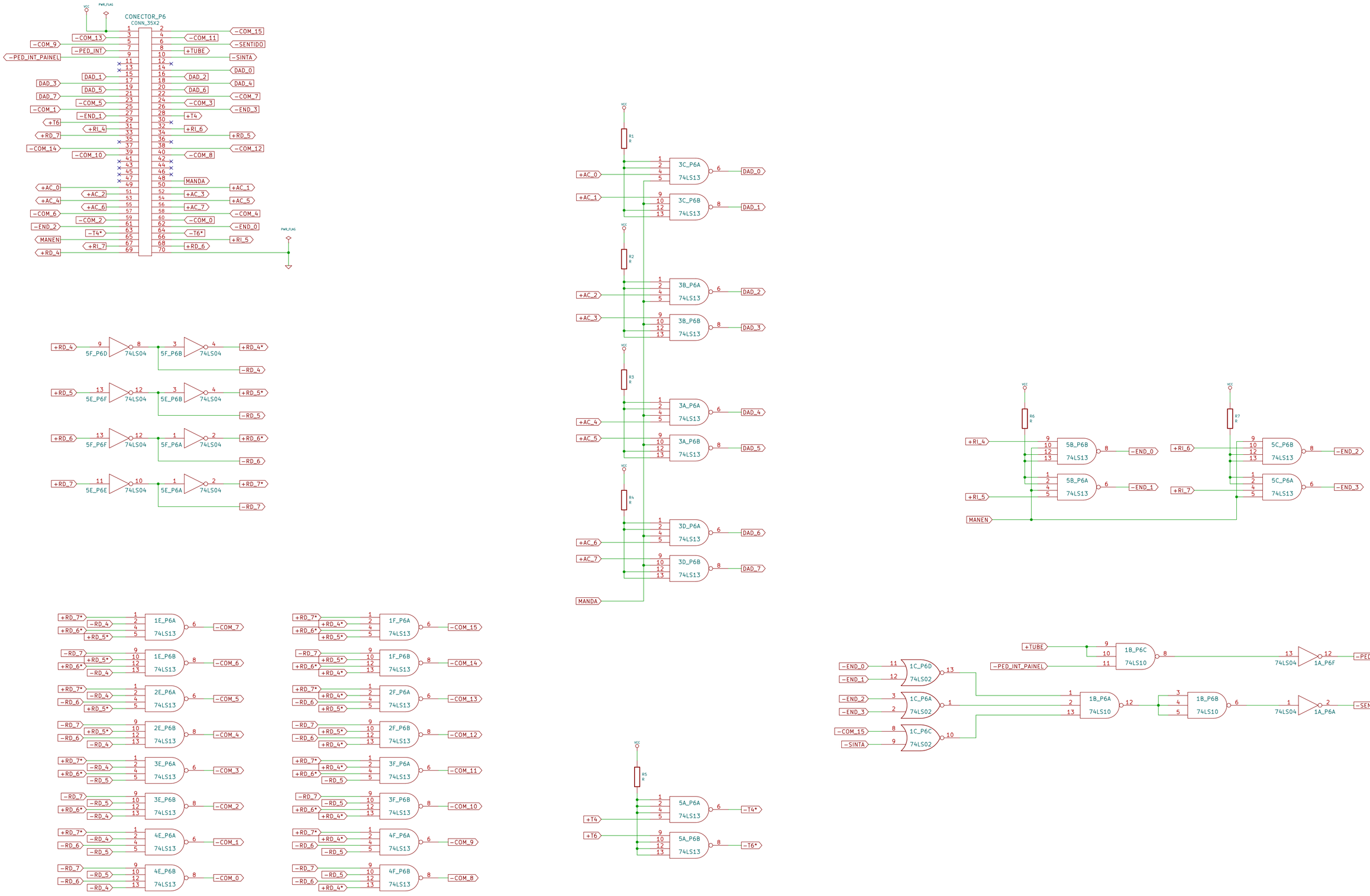




Sheet: /PRANCHA V: PLACA CONTROLE DA MEMÓRIA/
File: FCM–7.kicad_sch

Title:

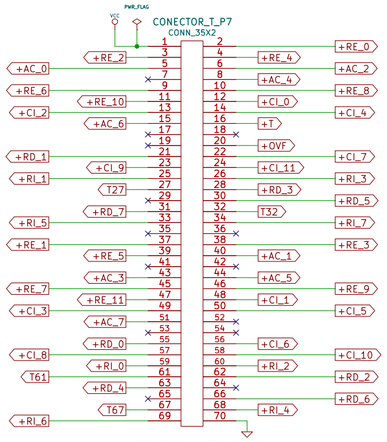
Size: A2	Date: 2 aug 2016	Rev:
KiCad E.D.A. 10.0.3		Id: 7/18



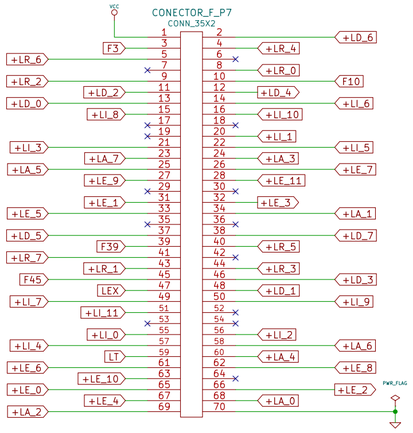
Sheet: /PRANCHA VI: DADOS PARA CARTÕES INTERFACE/
File: FIN-8.kicad_sch

Title:

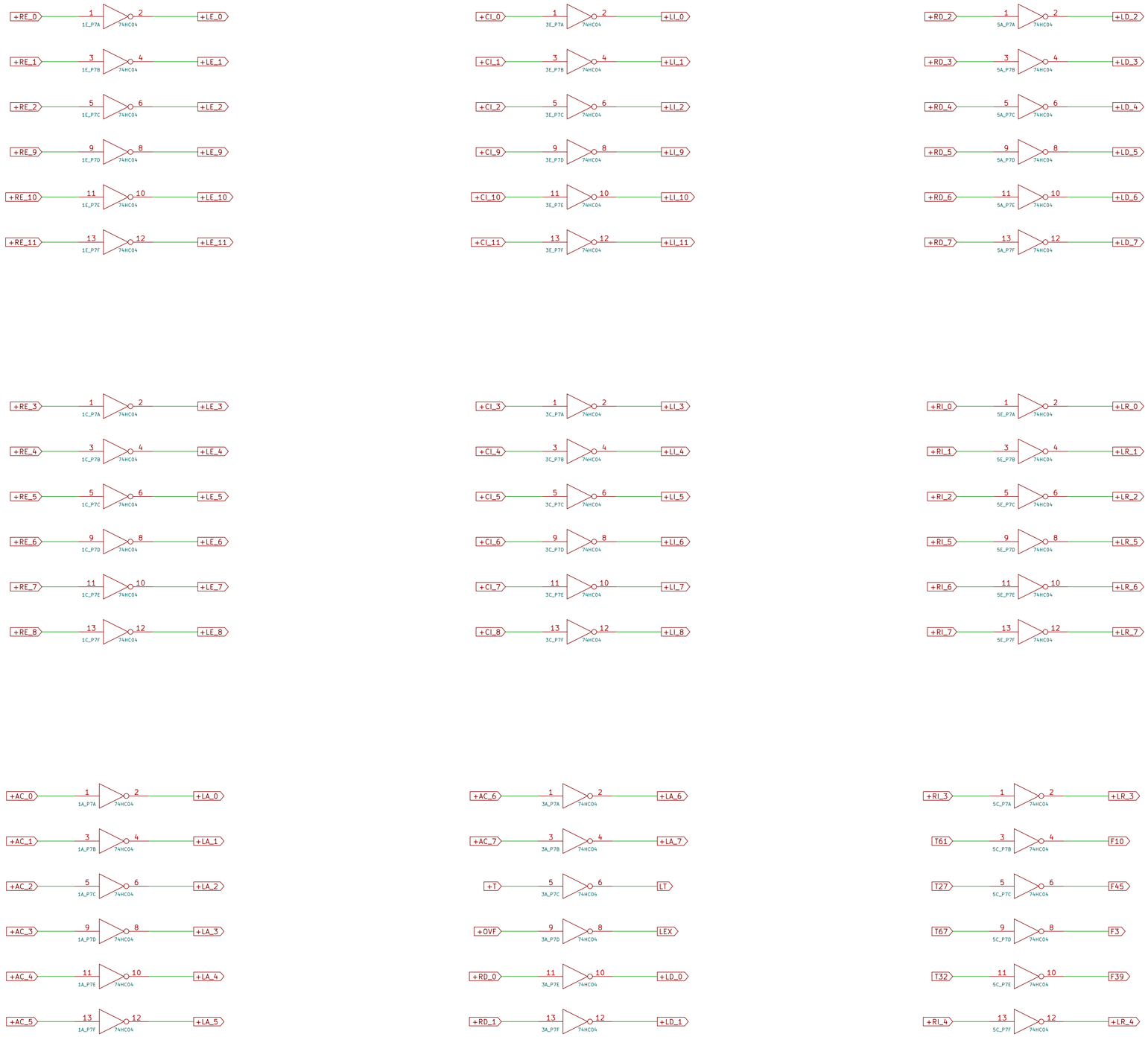
Size: A2	Date: 2 aug 2016	Rev:
KiCad E.D.A. 10.0.3		Id: 8/18



CONECTOR
TRASEIRO



CONECTOR
FRONTAL



Sheet: /PRANCHA VII: PLACA MOSTRA REGISTRADORES/
File: PMR-1.kicad_sch

Title:

Size: A2	Date: 2 aug 2016	Rev:
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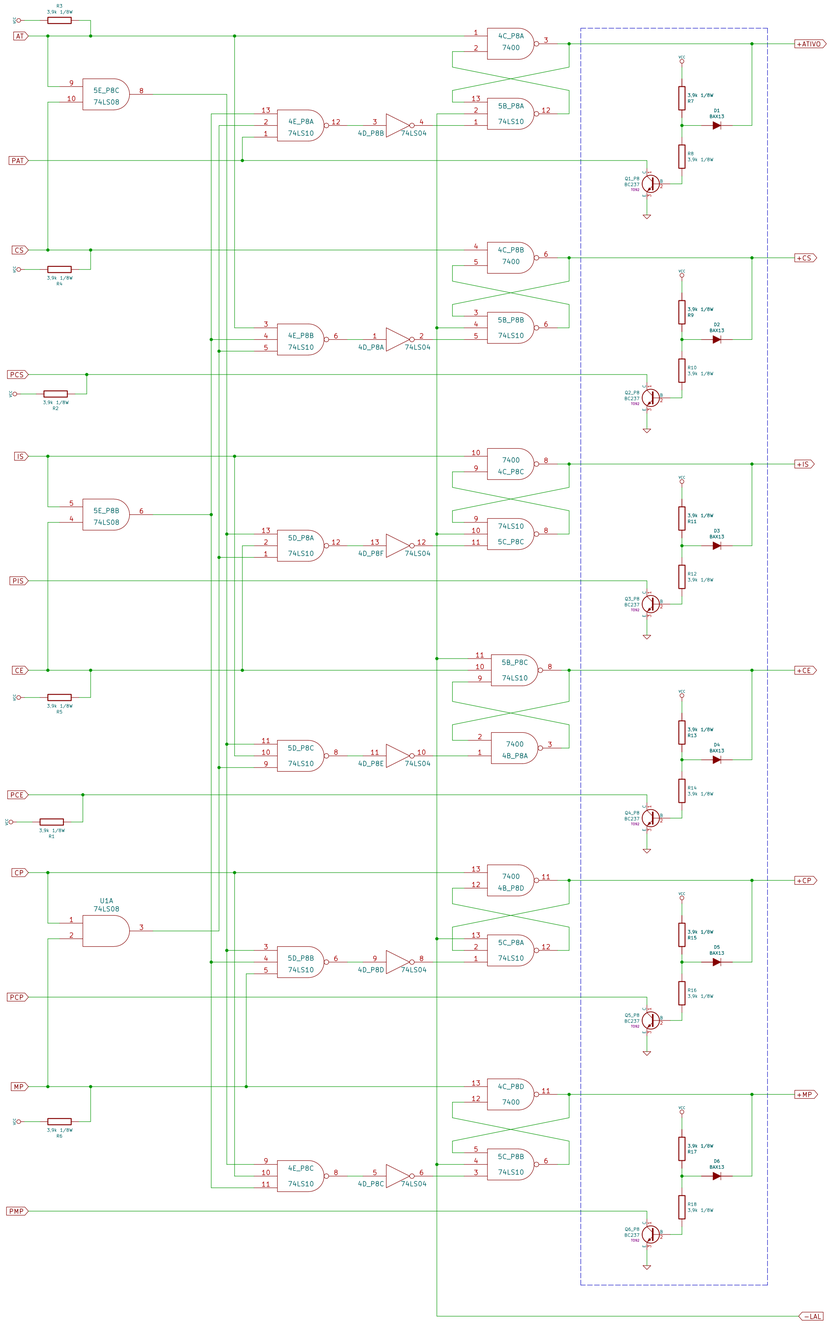
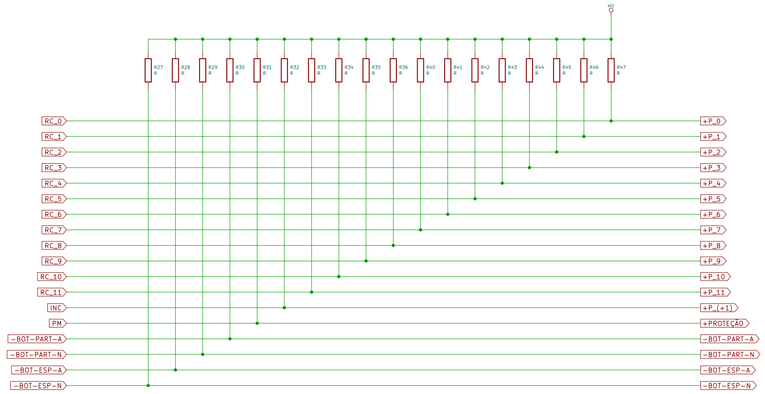
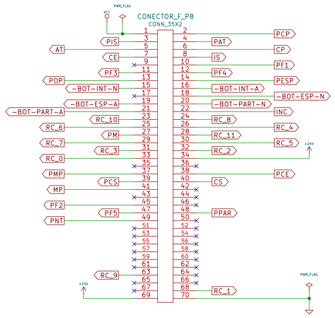
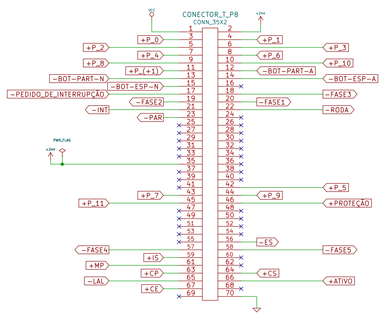


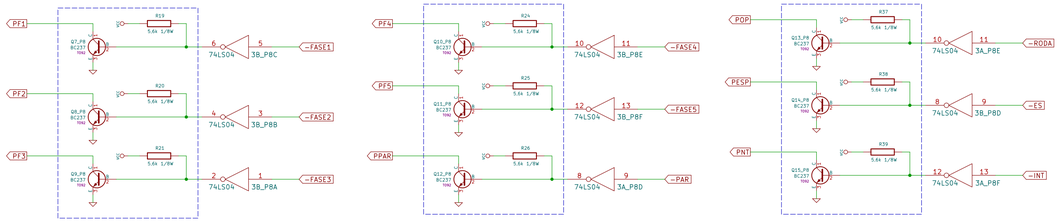


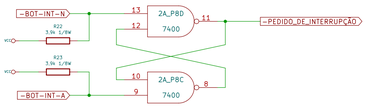


CONECTOR
FRONTAL



CONECTOR
TRASEIRO





(!) Isso aparenta estar errado.
Há ainda 2 unidades disponíveis na pastilha 4B.
Por que não usá-las aqui?

Sheet: /PRANCHA VIII: PLACA CHAVES E PILOTO/
File: PCP-2.kicad_sch

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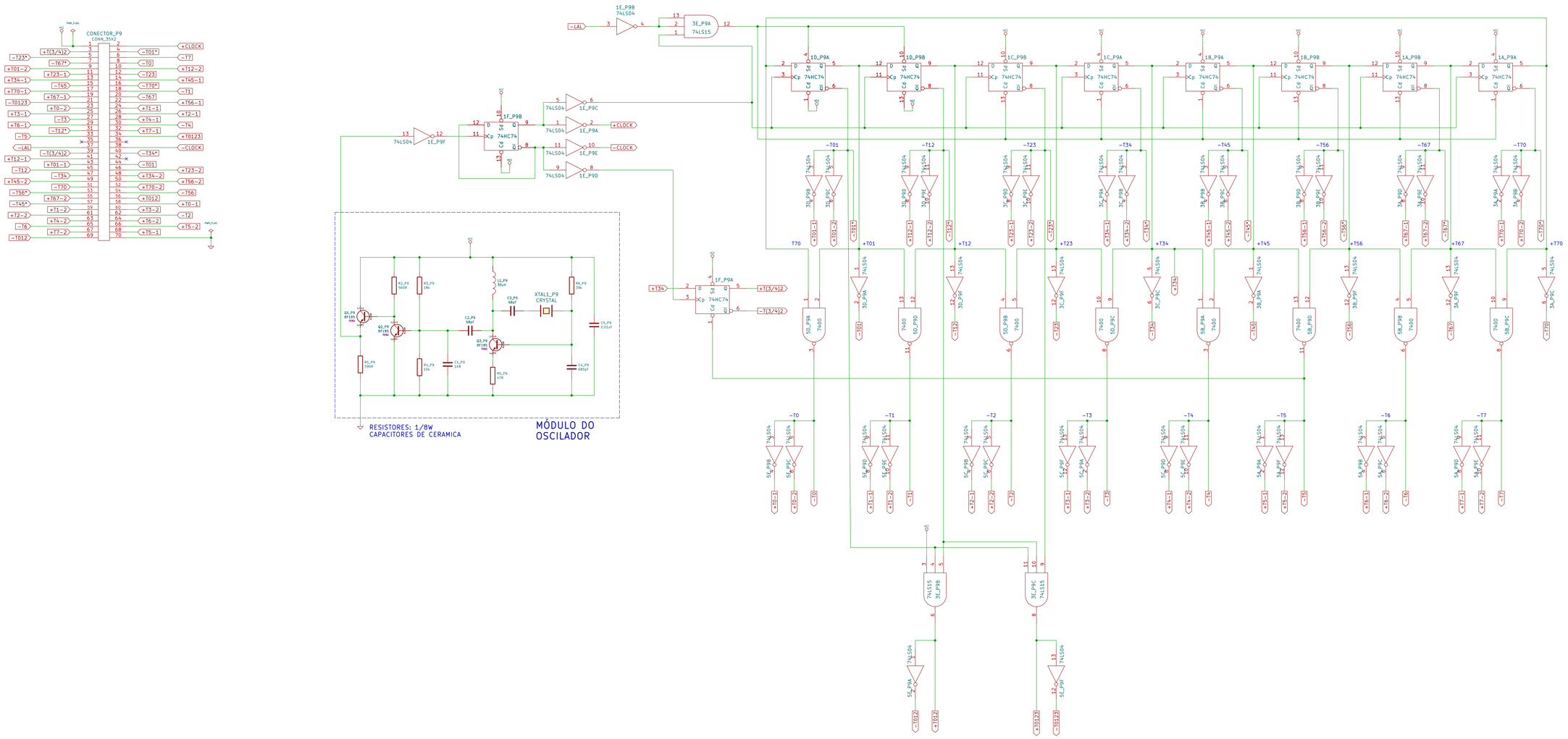
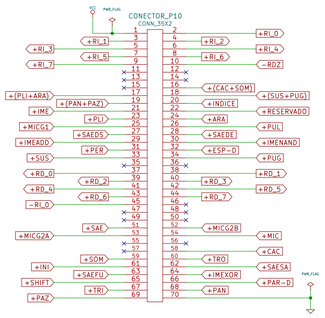
Size: A1	Date: 2 aug 2016	Rev:
KiCad E.D.A. 10.0.3		Id: 10/18

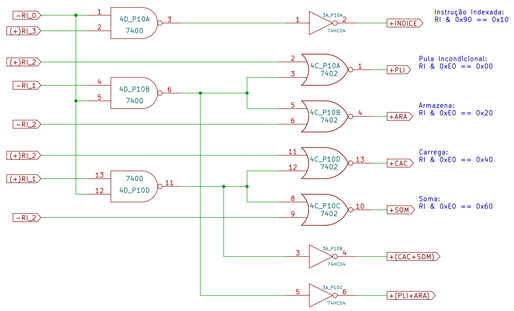


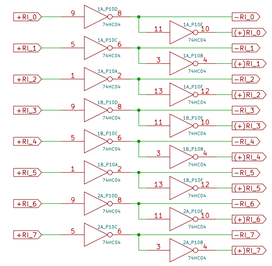
Sheet: /PRANCHA IX: RELÓGIO CENTRAL/
File: CRC-1.kicad_sch

Title:

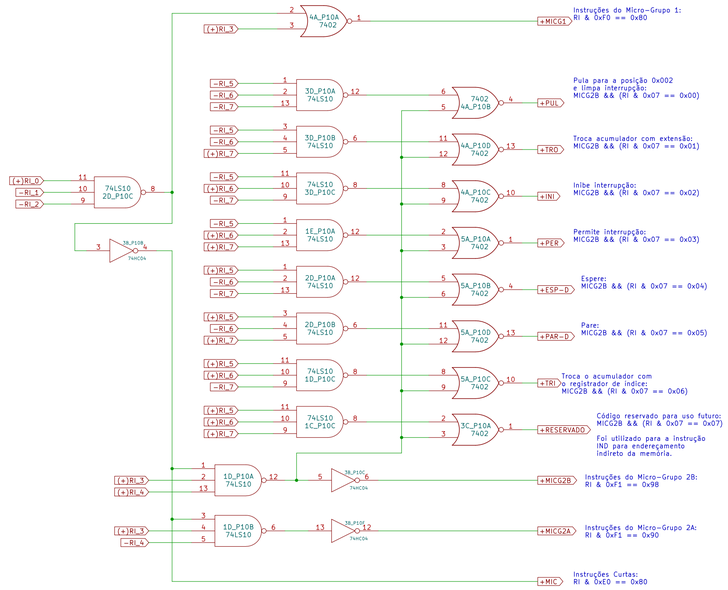
Size: A1 Date: 2 aug 2016	Rev:
KiCad E.D.A. 10.0.3	Id: 11/18

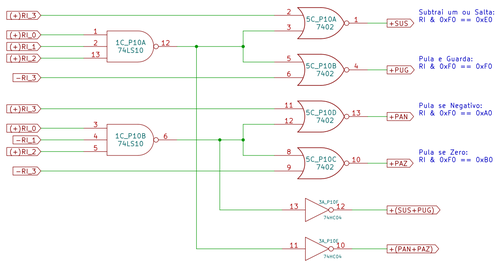


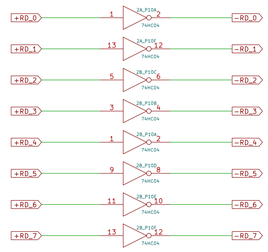


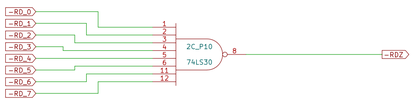


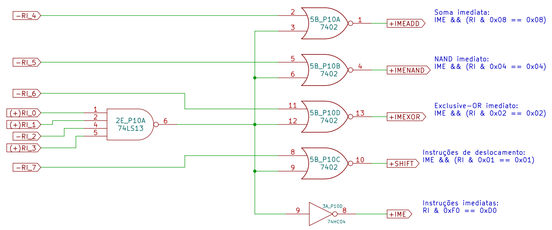
Observação:
Os códigos de 3 bits para estas instruções divergem dos valores listados na tabela 5-6 de FREGNI/LANGDON

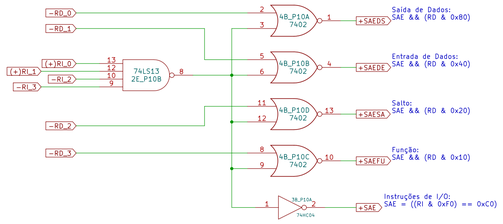


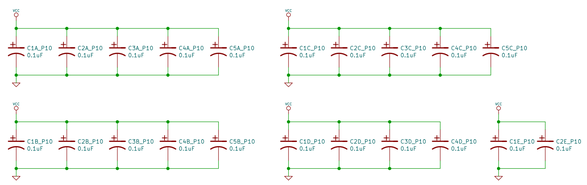




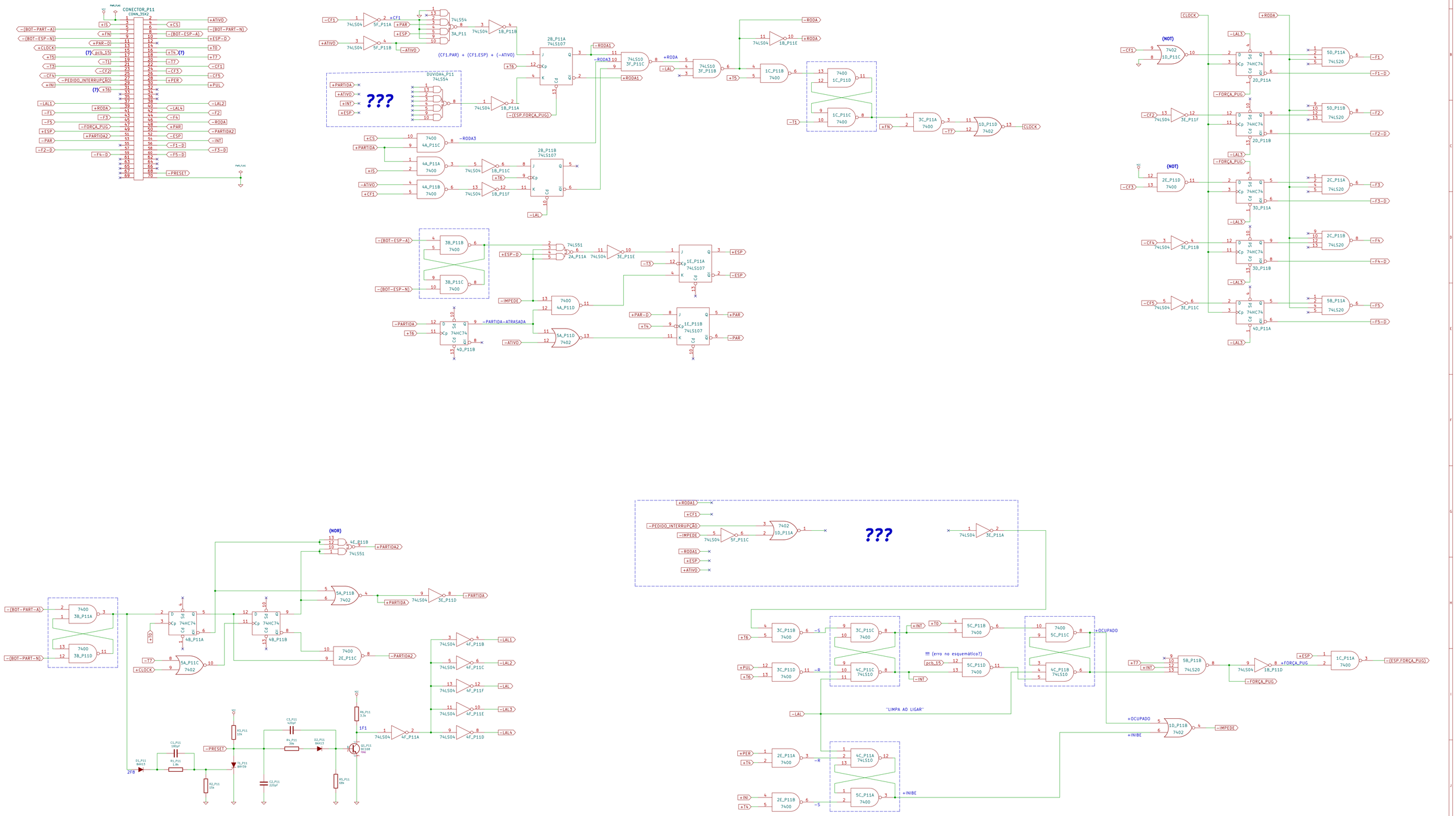








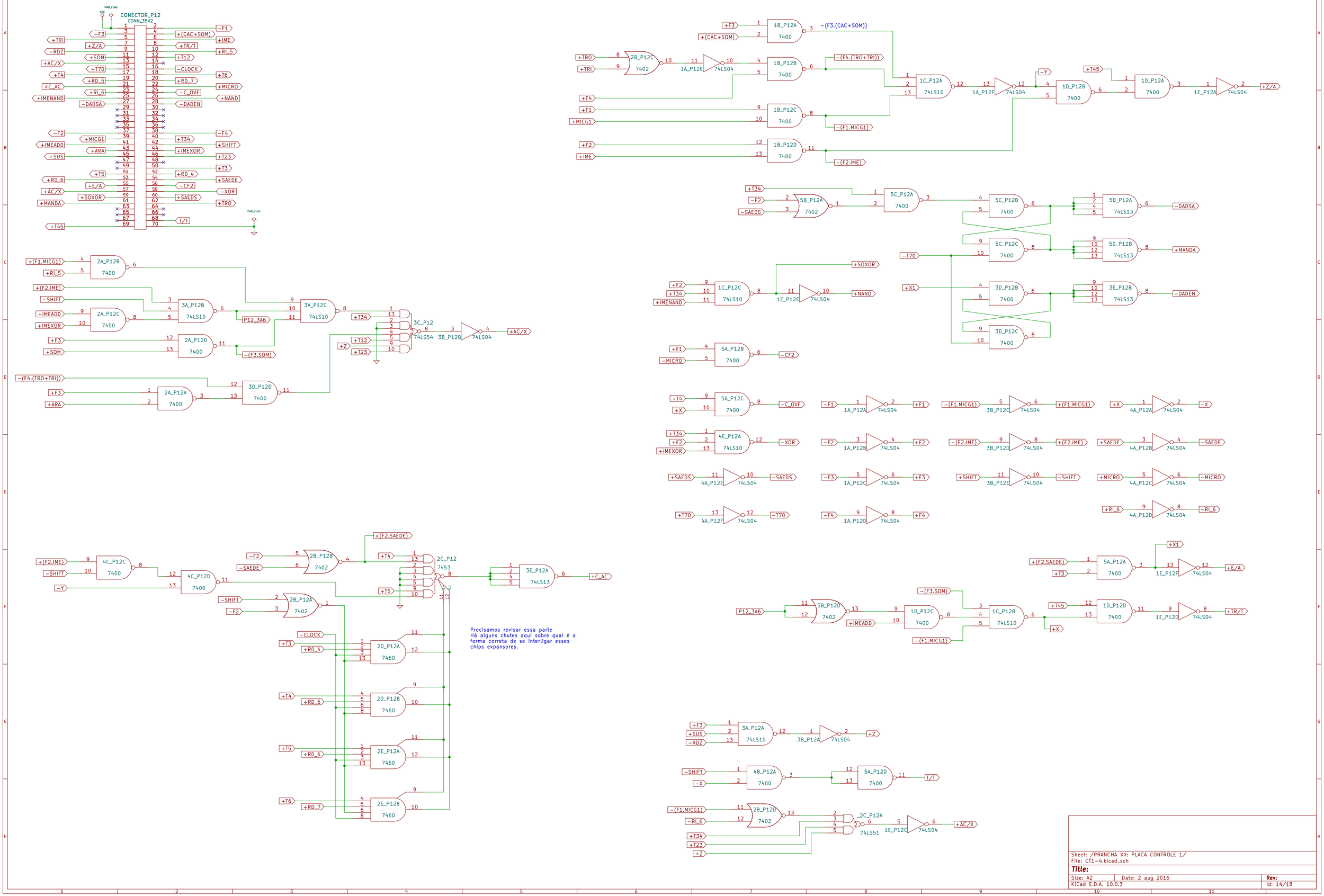
Sheet: /PRANCHA X: DECODIFICADOR/		
File: CDE-3.kicad_sch		
Title:		
Size: A1	Date: 2 aug 2016	**Rev:**
KiCad E.D.A. 10.0.3		Id: 12/18



Sheet: /PRANCHA XI: CONTROLE DE ESTADO/
File: CCE-2.kicad_sch

Title:

Size: A1	Date: 2 aug 2016	Rev:
KiCad E.D.A. 10.0.3		Id: 13/18



Sheet: /PRANCHA XII: PLACA CONTROLE 1/
File: CT1-4.kicad_sch

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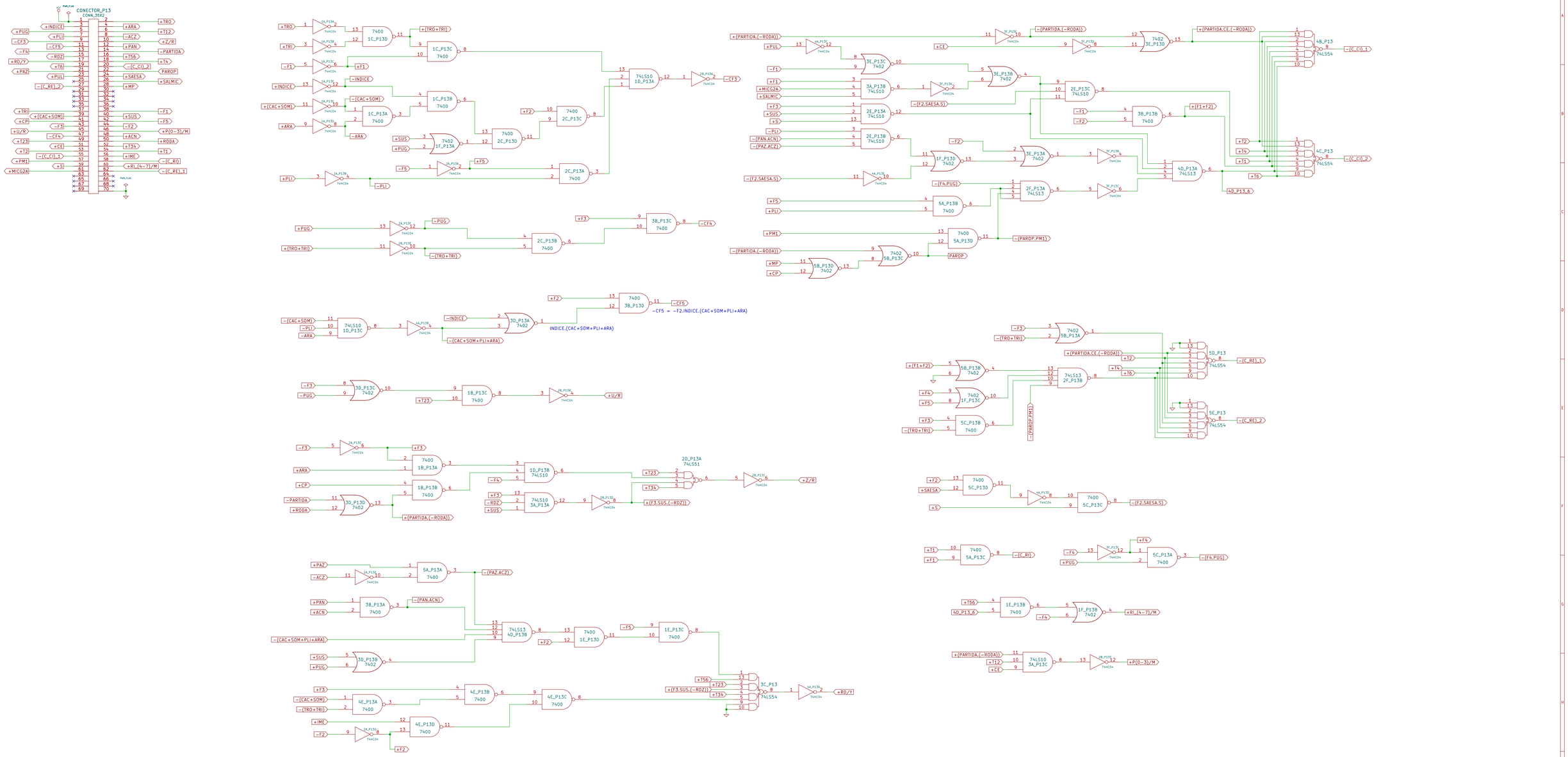
KiCad E.D.A. 10.0.3		Id: 14/18



Sheet: /PRANCHA XIII: PLACA SINAIS CONTROLE/
File: CT2-5.kicad_sch

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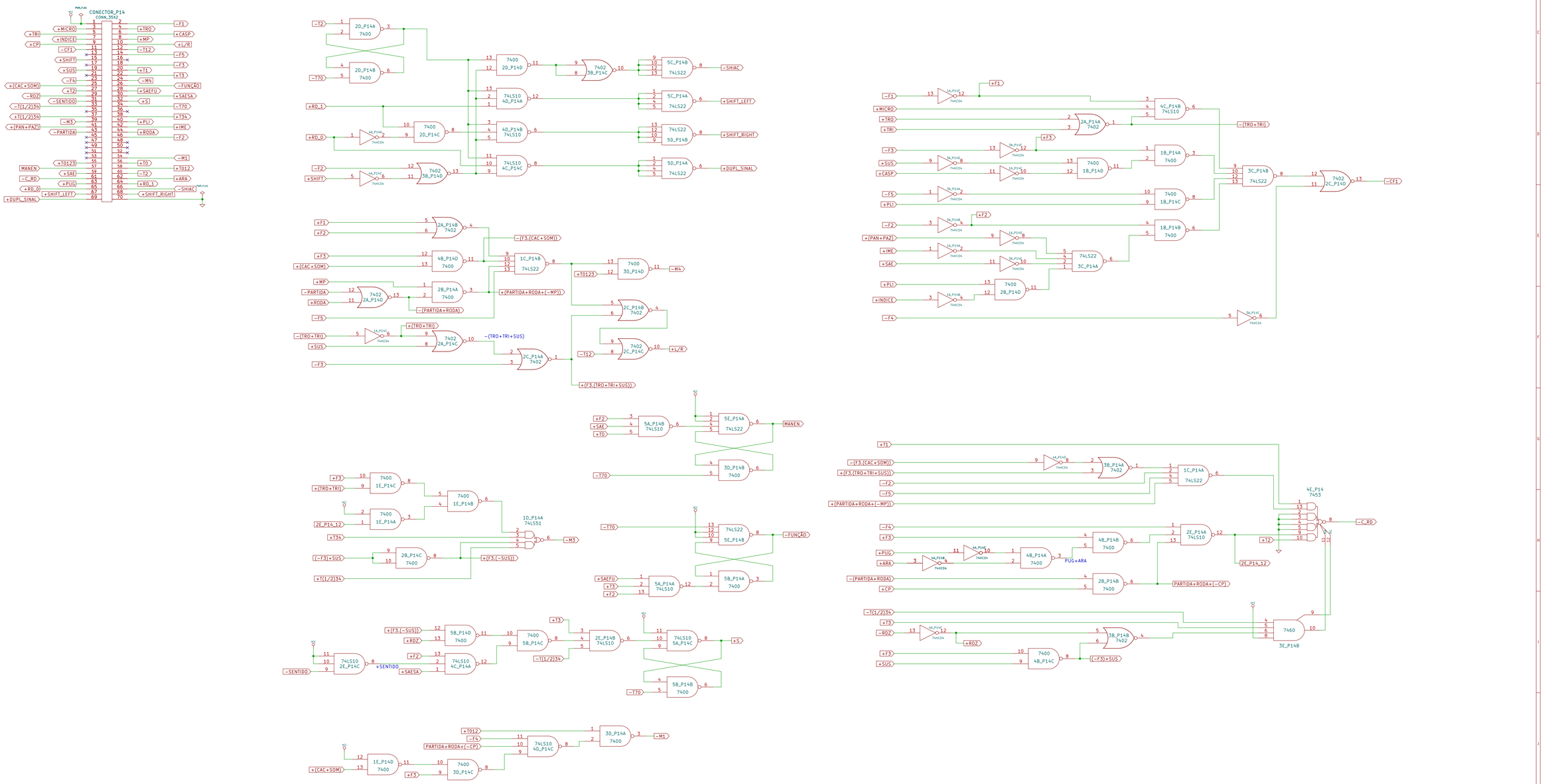
Size: A1	Date: 2 aug 2016	Rev:
KiCad E.D.A. 10.0.3		Id: 15/18



Sheet: /PRANCHA XIV: PLACA DE CONTROLE/	
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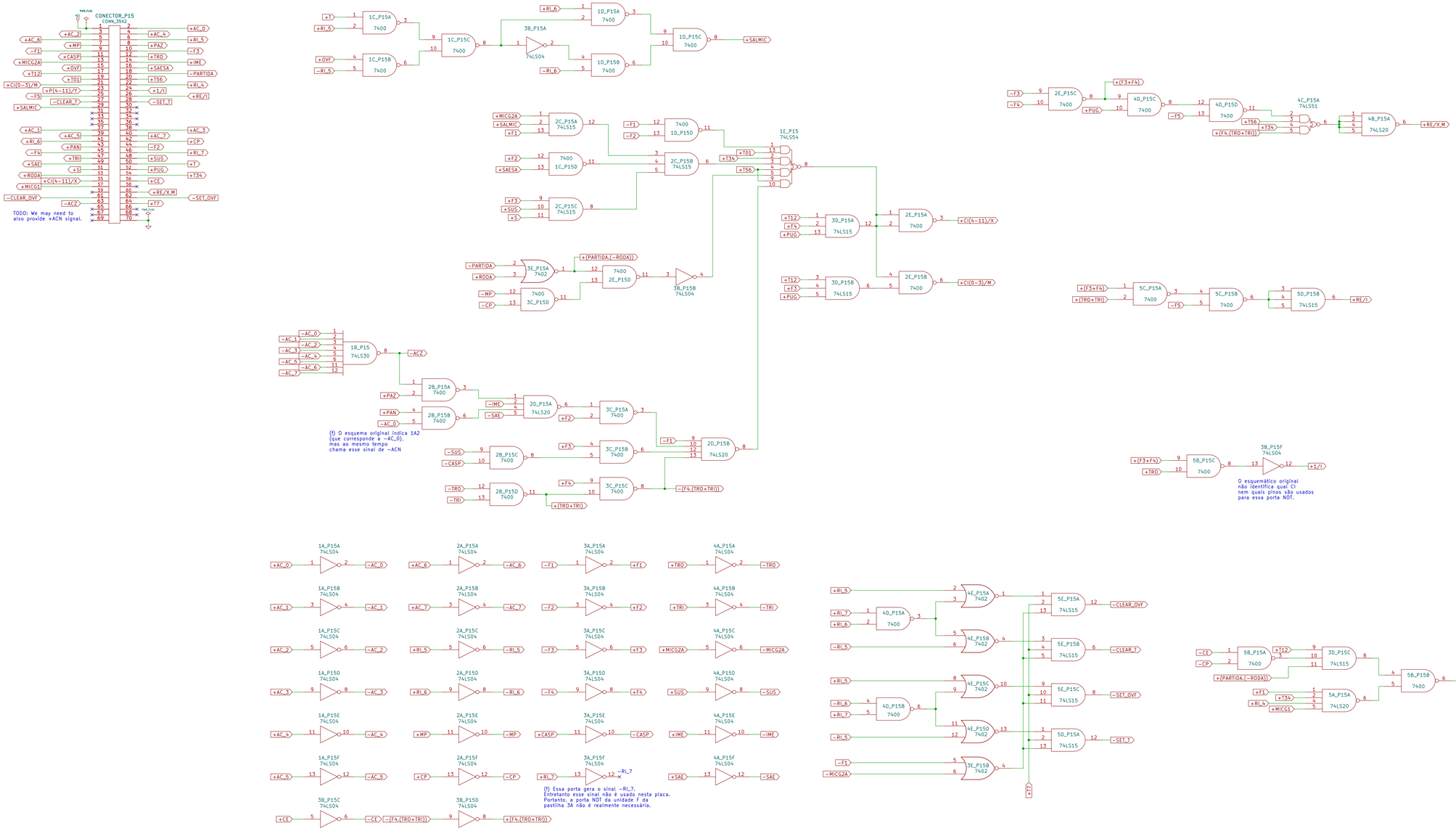
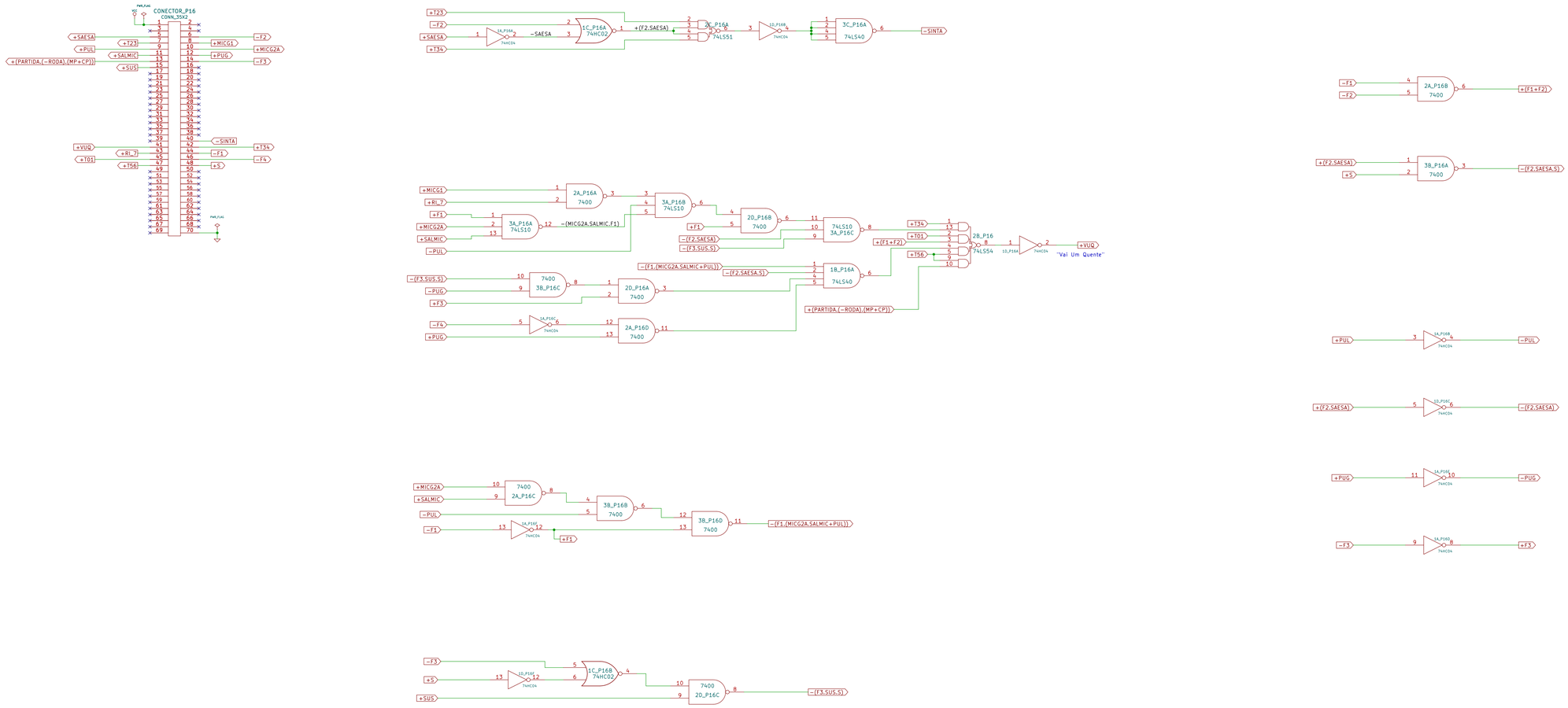
Title:	
Size: A1	Date: 2 aug 2016
KiCad E.D.A. 10.0.3	Id: 16/18



Sheet: /PRANCHA XV: SINAIS DE CONTROLE/	
File: CPA-7.kicad_sch	
Title:	
Size: A1 Date: 2 aug 2016	Rev:
KiCad E.D.A. 10.0.3	Id: 17/18



Re-desenho: Felipe Sanches
Projeto: Edson Fregni
Sinais de Controle
Computador de Controle - Patinho Feio
EPUSP LSD
Sheet: /PRANCHA XVI: SINAIS DE CONTROLE/
File: CT5-B.kicad_sch

Title: *PRANCHA XVI*

Size: A1 | Date: 2 aug 2016	Rev:
KICad E.D.A. 10.0.3	Id: 18/18